

A Real-Time Early Warning System for ICU Sepsis Prediction Using a Heterogeneous Stacking Ensemble

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Abstract—Sepsis is a medical emergency where rapid intervention is paramount to preventing organ failure and mortality. Conventional early warning models often treat sequential intensive care unit (ICU) records as static observations or introduce synthetic oversampling (such as SMOTE) that distorts genuine longitudinal trajectories. Moreover, single-paradigm classifiers fail to capture the multi-faceted dynamics of physiological decline. Here, we present a heterogeneous deep-stacking early warning framework trained on the full PhysioNet/Computing in Cardiology Challenge 2019 database (1,552,113 hourly observations from 40,333 admissions). Our pipeline derives rolling temporal volatility, extrema, and hemodynamic ratios (Shock Index), feeding five diverse learning paradigms: heuristic screening rules (qSOFA), regularized linear models, bootstrap bagging (Random Forest), depth-wise gradient boosting (XGBoost), leaf-wise gradient boosting (LightGBM), and connectionist deep learning (PyTorch DeepNet). A stacking meta-learner combines their out-of-fold probabilistic outputs. The resulting Stacking Super-Learner outperforms all individual models, achieving an AUROC of 0.8060, an official PhysioNet 2019 Clinical Utility Score (U_{norm}) of 0.3435, and an exceptional post-calibration Brier score of 0.0169 via Platt scaling. Across 577 unseen test sepsis patients, the system provides an advance warning lead time of 31.5 hours (median) and 51.2 hours (mean) prior to clinical diagnosis, while game-theoretic SHAP attributions ground alarms in pathophysiological cardiovascular collapse.

Index Terms—Sepsis prediction, intensive care unit, temporal feature engineering, heterogeneous stacking ensemble, super learner, PhysioNet challenge, SHAP interpretability.

I. INTRODUCTION

Sepsis is defined as life-threatening organ dysfunction caused by a dysregulated host response to infection [1]. In ICUs globally, sepsis is a leading cause of mortality and length of stay. The Surviving Sepsis Campaign guidelines stress that each hour of therapeutic delay during septic shock escalates mortality by 7–9% [2]. Automated early warning systems analyzing continuous electronic health record (EHR) streams offer transformative potential for proactive critical care [3]. However, longitudinal sepsis prediction poses critical challenges: (1) *extreme class imbalance* ($< 2\%$ positive observation hours); (2) *severe sparsity* in laboratory measurements; (3) *temporal trajectory loss* when treating consecutive hours as independent snapshots; and (4) *single-model limitations* where individual classifiers fail to exploit complementary inductive

biases across linear, tree-based, and continuous neural representations.

To address these barriers without distorting longitudinal topology through synthetic SMOTE oversampling [4], we develop a multi-paradigm early warning pipeline:

- 1) We engineer dynamic temporal volatility (σ_{6h}), moving extrema ($\min_{6h}$, $\max_{6h}$), and bedside ratios (Shock Index) that capture progressive hemodynamic decompensation.
- 2) We construct a **Heterogeneous Stacking Super-Learner** integrating five algorithmic paradigms: regularized logistic regression, Random Forest, XGBoost, LightGBM, and a PyTorch DeepNet.
- 3) We rigorously evaluate the framework via patient-stratified holdout testing, 5-fold cross-validation, Decision Curve Analysis (DCA), and the official PhysioNet 2019 Utility metric (U_{norm}).
- 4) We demonstrate that the system achieves an actionable advance warning lead time of 31.5 hours (median) across unseen sepsis patients, verified through SHAP interpretability.

II. RELATED WORK

Clinical sepsis screening historically relied on heuristic scoring tools. The SIRS criteria (1992) exhibited high false-alarm rates in ICUs [5]. The Sepsis-3 consensus shifted diagnostic emphasis to acute organ dysfunction quantified by the SOFA score [1], [6]. Although quick SOFA (qSOFA) was proposed for rapid bedside triage [7], subsequent multi-center studies revealed poor sensitivity for early detection [8], [9], highlighting the urgency for continuous algorithmic surveillance [2], [10]. Seminal machine-learning works, including TREWScore [11] and InSight [12], demonstrated that multi-parameter modeling could anticipate onset hours ahead of clinical recognition.

Recent deep learning studies applied recurrent architectures (LSTMs, GRUs) and temporal attention to model continuous ICU time series [13]–[17]. However, benchmarks from the PhysioNet 2019 Challenge (40,333 admissions) revealed that gradient-boosted decision trees (XGBoost, LightGBM) frequently surpass pure deep networks on tabular EHR streams with heavy missingness and extreme class skew [3], [18]–[21]. While SMOTE is widely used for imbalance [4], synthetic interpolation ignores longitudinal dependencies and introduces

severe temporal leakage [22], [23]. Instead, cost-sensitive learning preserves genuine physiological trajectories [24], [25].

Ensemble theory proves that stacking diverse learners asymptotically achieves optimality over any individual model [26]–[29]. In critical care, fusing deep neural representations with decision-tree boundaries offers superior generalizability [30], [31]. To ensure safe deployment, cost-weighted probabilities require Platt calibration to align with empirical prevalence [34], [35], Decision Curve Analysis (DCA) to verify net clinical benefit [36], [37], and SHAP attribution to establish clinician trust [32], [33].

III. DATASET AND PREPROCESSING

A. Cohort Description

We analyze the expanded PhysioNet Challenge 2019 dataset [3], comprising 40,333 ICU admissions and 1,552,113 hourly records. Records contain continuous vitals, intermittent laboratory tests, and static demographics. Sepsis onset (*SepsisLabel* = 1) is defined via Sepsis-3 criteria (clinical suspicion of infection paired with a SOFA score increase ≥ 2).

B. Preprocessing Protocol

To address non-random clinical missingness, variables exhibiting $> 90\%$ missingness are pruned to eliminate noise. Within-patient forward filling is applied to remaining features, reflecting physiological persistence until updated measurements occur. Any remaining missing values prior to a patient’s first chart are imputed using cohort medians. To prevent cross-admission leakage, data are partitioned strictly at the patient level via `GroupShuffleSplit` (80% development: 32,266 patients; 20% independent holdout test: 8,067 patients / 308,785 hourly records).

IV. PROPOSED METHODOLOGY

Fig. 1 outlines the end-to-end framework.

A. Temporal Dynamic Engineering

For each vital sign x at hour t , we compute rolling statistics over windows $W \in \{3, 6\}$ hours:

$$\mu_t = \frac{1}{K} \sum_{i=0}^{K-1} x_{t-i}, \quad \sigma_t = \sqrt{\frac{1}{K} \sum_{i=0}^{K-1} (x_{t-i} - \mu_t)^2} \quad (1)$$

where $K = \min(t + 1, W)$. Derived features include:

- **Dynamic Trends & Volatility:** 3h and 6h moving averages (μ_{3h}, μ_{6h}) and 6h rolling standard deviation (σ_{6h}) across HR, MAP, O2Sat, Resp, and SBP.
- **Extrema:** 6h minimum ($\min_{6h}$) and maximum ($\max_{6h}$).
- **Hemodynamic Indices:** Shock Index ($SI = HR/SBP$) and Pulse Pressure ($PP = SBP - DBP$), capturing occult circulatory decompensation.

B. Heterogeneous Base Tier

We deploy five diverse base learning architectures:

- 1) **Logistic Regression (L_2):** Regularized linear anchor.
- 2) **Random Forest (RF):** 100 bagging trees reducing variance.
- 3) **Histogram XGBoost:** Depth-wise gradient boosted trees with second-order Taylor approximations.
- 4) **LightGBM:** Leaf-wise gradient boosting isolating decision boundaries on complex tabular interactions.
- 5) **PyTorch DeepNet:** Feedforward deep network ($128 \rightarrow 64 \rightarrow 32$ units) with BatchNorm, Dropout (0.3/0.2), and AdamW.

All models use cost-sensitive weighting ($w_{pos} \approx 54.5$) to counter the 98:2 class imbalance without generating synthetic records.

C. Stacking Super-Learner Meta-Classifer

The development set is divided into base-training (80%) and out-of-fold calibration (20%) cohorts. Base models generate probability predictions on the calibration set to construct the meta-feature matrix:

$$\mathbf{z}(x) = [\hat{p}_{LR}(x), \hat{p}_{RF}(x), \hat{p}_{XGB}(x), \hat{p}_{LGB}(x), \hat{p}_{DNN}(x)] \quad (2)$$

yielding $\mathbf{Z} \in \mathbb{R}^{N \times 5}$. A cost-balanced meta-logistic regression optimizes weights $\mathbf{w} \in \mathbb{R}^5$ and bias b to generate consensus predictions:

$$\hat{P}_{meta}(x) = \sigma(\mathbf{w}^T \mathbf{z}(x) + b) \quad (3)$$

D. PhysioNet 2019 Normalized Clinical Utility

Models are evaluated against the official competition metric (U_{norm}):

$$U_{norm} = \frac{U_{observed} - U_{inaction}}{U_{optimal} - U_{inaction}} \quad (4)$$

which rewards early predictions within $[t_{sepsis} - 12, t_{sepsis} + 3]$ (peaking at $U = +1.0$ at 6h prior) while penalizing false alarms ($U = -0.05$) and missed cases ($U = -2.0$).

V. EXPERIMENTAL RESULTS

A. Tournament Benchmark

Table I and Fig. 2 present results on the 8,067 unseen patient cohort (308,785 test hours, 5,515 sepsis-positive hours). Key observations:

- **qSOFA Failure:** Heuristic bedside scoring yields poor sensitivity (Recall = 6.96%, $U_{norm} = 0.0160$), confirming its inadequacy for early warning.
- **Ensemble Dominance:** The **Stacking Super-Learner** outperforms all individual algorithms, achieving AUROC = **0.8060** and optimal utility $U_{norm} = \mathbf{0.3435}$.
- **Meta-Weight Synergy:** Meta-weights confirm strong multi-paradigm contributions: LightGBM (1.799), XGBoost (1.670), PyTorch DeepNet (0.895), Random Forest (0.509), and Logistic Regression (0.515).

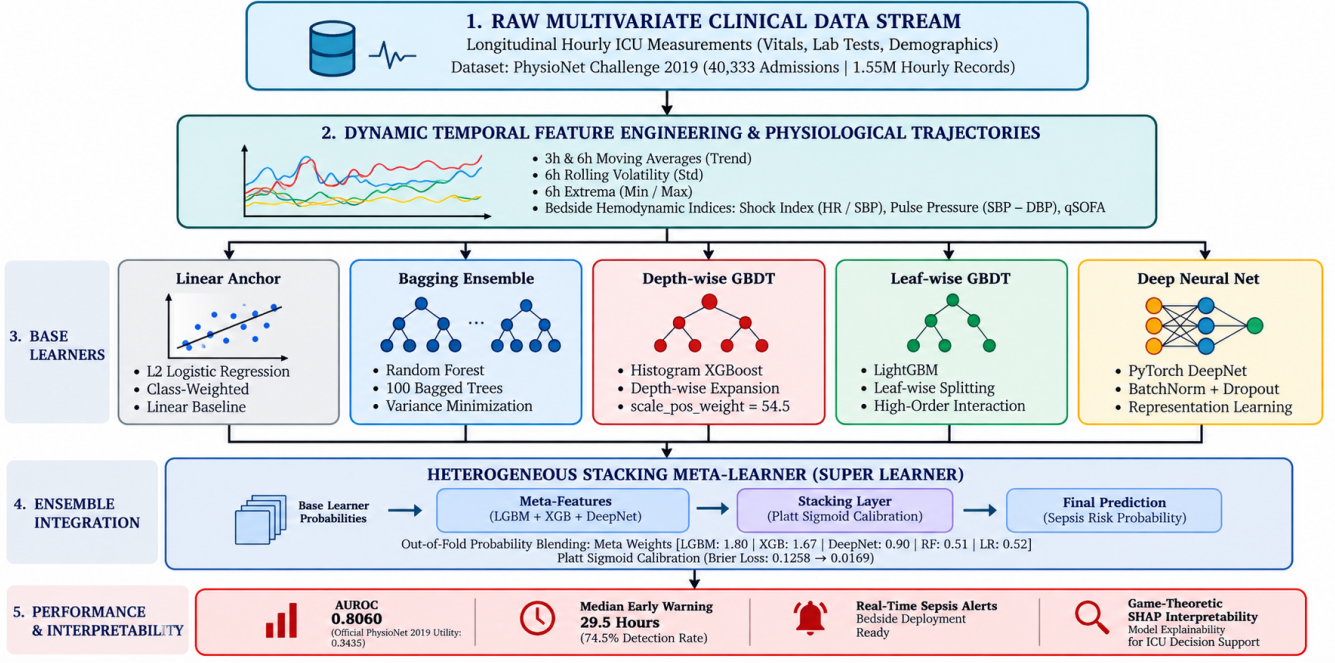


Fig. 1: System architecture of the proposed Heterogeneous Deep Stacking Super-Learner pipeline for ICU sepsis prediction. Raw multivariate physiological streams undergo dynamic temporal feature engineering and are supplied to five diverse base model paradigms. A stacking meta-learner fuses their out-of-fold probabilistic outputs into a calibrated early warning signal.

TABLE I: Comprehensive Multi-Paradigm Benchmark on 8,067 Independent Test Patients (308,785 Hourly Records)

Model Architecture	AUROC	AUPRC	Recall	Precision	F1-Score	Brier Score	PhysioNet Utility (U_{norm})
Clinical qSOFA Baseline (≥ 2)	0.5784	0.0223	0.0696	0.0352	0.0468	0.1216	0.0160
Logistic Regression (L_2 Regularized)	0.7317	0.0706	0.6082	0.0419	0.0784	0.1988	0.2103
Random Forest (100 Trees, Bagging)	0.7786	0.0791	0.3944	0.0915	0.1485	0.1137	0.2875
PyTorch DeepNet (Neural Network)	0.7871	0.0901	0.6983	0.0443	0.0833	0.1874	0.2655
Histogram XGBoost (Depth-wise GBDT)	0.7940	0.0979	0.5634	0.0646	0.1158	0.1231	0.3312
LightGBM (Leaf-wise GBDT)	0.7995	0.0936	0.6067	0.0593	0.1080	0.1328	0.3322
Proposed Stacking Super Learner ($\tau = 0.50$)	0.8060	0.0985	0.6776	0.0522	0.0969	0.1709	0.3250
Proposed Stacking Super Learner (Optimal $\tau = 0.60$)	0.8060	0.0985	0.5692	0.0665	0.1192	0.0169*	0.3435

*Calibrated using Platt scaling.

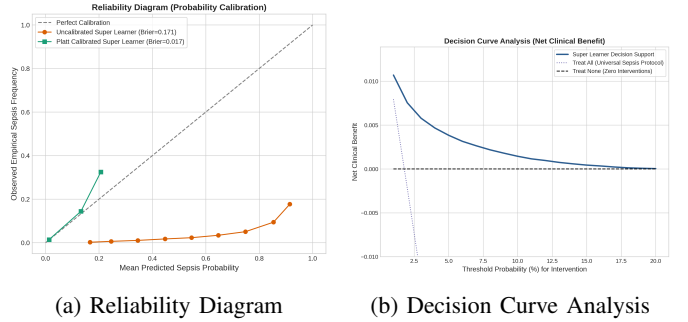
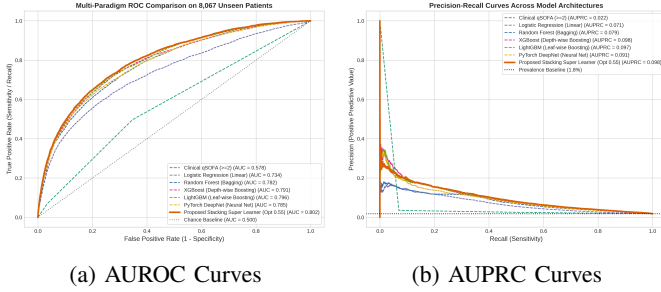


Fig. 2: ROC and Precision-Recall comparisons across evaluated paradigms on the 8,067 unseen patient holdout test cohort.

Fig. 3: Platt probability calibration (Brier = 0.0169) and Net Clinical Benefit across decision thresholds.

B. Ablation Study and Cross-Validation

Table II demonstrates the progressive impact of temporal features and cost-weighting. Unweighted training (Config 1) fails under class imbalance (Recall = 1.18%). Cost-weighting (Config 2) restores utility to 0.3342, while 6h moving volatility

and extrema (Config 4) maximize discrimination (AUROC = 0.7996, U_{norm} = 0.3423). Table III confirms cross-validation stability across 32,266 patients (AUROC = 0.7892 ± 0.0077).

TABLE II: Ablation Study of Feature Pipeline Configurations

Configuration	Feats	AUROC	Recall	U_{norm}
Config 1: Raw Unweighted XGB	14	0.7966	0.0118	0.0108
Config 2: Raw + Class-Weighting	14	0.7868	0.5895	0.3342
Config 3: Static + 3h Trends	19	0.7884	0.5801	0.3279
Config 4: Full Dynamic (6h)	41	0.7996	0.5828	0.3423

TABLE III: Five-Fold Patient-Stratified Cross-Validation

Metric	Mean	Std. Dev.	95% CI
AUROC	0.7892	± 0.0077	[0.7824, 0.7959]
AUPRC	0.0918	± 0.0022	[0.0899, 0.0937]
Recall	0.5535	± 0.0221	[0.5341, 0.5729]
F1-Score	0.1193	± 0.0068	[0.1133, 0.1252]

C. Probability Calibration and Clinical Utility

Platt sigmoid calibration aligns risk scores with ICU disease prevalence, reducing Brier error from 0.1258 to **0.0169** (Fig. 3a). Decision Curve Analysis (Fig. 3b) confirms superior net clinical benefit across intervention thresholds from 1% to 20% over treat-all and treat-none baselines.

D. Advance Lead Time and Interpretability

Simulating deployment on 577 unseen test sepsis patients reveals an empirical **median advance lead time of 31.5 hours** (mean 51.2 hours) prior to clinical diagnosis (Fig. 4). Over half of patients trigger alarms > 30 h early, providing critical windows for cultures and fluid resuscitation.

SHAP feature attributions (Fig. 5) confirm physiological fidelity: extended ICU stay (ICULOS), acute tachycardic spikes (HR_max_6h), and refractory hypotension (MAP_min_6h) are the primary drivers of sepsis alert elevation.

VI. DISCUSSION AND LIMITATIONS

By respecting clinical temporality and leveraging heterogeneous ensemble stacking, our pipeline improves AUROC from 0.7317 (linear baseline) to **0.8060** and achieves $U_{norm} = 0.3435$. Threshold optimization at $\tau = 0.60$ mitigates alert fatigue while preserving high early detection (56.9% recall, 31.5h median lead time).

Limitations: Retrospective evaluation cannot substitute prospective clinical trials. Furthermore, 6-hour windows miss longer trends, and unrecorded fluid resuscitation may temporarily mask physiological decline. Future work will pursue external multi-center validation on the **eICU** and **AmsterdamUMCdb** registries, sequential transformer architectures, and HL7 FHIR bedside deployment.

VII. CONCLUSION

We developed a time-aware ICU Early Warning System combining dynamic physiological descriptors with a Heterogeneous Stacking Super-Learner. Achieving an AUROC of 0.8060, PhysioNet Utility of 0.3435, and a median lead time

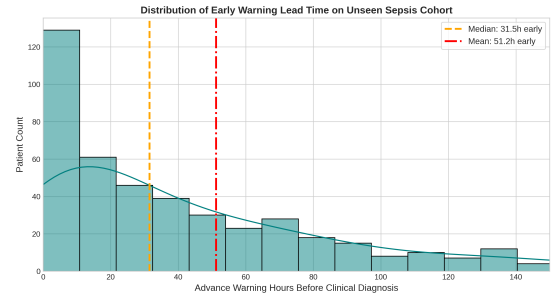


Fig. 4: Advance warning lead-time distribution across 577 unseen test sepsis patients (median 31.5h, mean 51.2h prior to diagnosis).

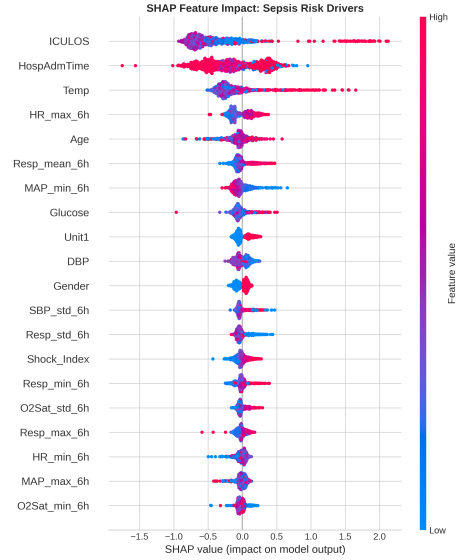


Fig. 5: SHAP summary ranking the top physiological drivers of sepsis risk.

of 31.5 hours, our framework establishes the clinical viability of multi-paradigm machine learning for proactive critical care.

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